Failure Analysis Report

FAIL-0617 — Assertion Failure

1. Failure Summary

Failure ID	FAIL-0617
Failure Type	Assertion Failure
Severity	Trivial
Component	Sensor Manager
Application	APP-006 — XRPlatform
Build Version	5.2.7
Timestamp	2024-07-21 09:50:10
Environment	ENV-003 — Hub Lab
OS Version	Tizen 7.0
Network	WiFi-6
Resolution	In Progress

2. Test Case Reference

Test Case ID	TC-2024
Title	Confirm that the system recovers gracefully after Data Mismatch during Bluetooth...
Execution ID	EXEC-03143
Test Run	RUN-006 — Regression Cycle 6 — Build 5.2.1
Requirement	REQ-0112 — Error Handling — Background Sync

3. Expected vs Observed Behaviour

Expected Behaviour	The system should complete firmware over-the-air update within the defined SLA.
Observed Behaviour	The system exhibited assertion failure when Bluetooth interference is present.

4. Error Details

Error Message:

```
AssertionError: expected '200' but received '404'.
```

Stack Trace / Log Excerpt:

```
com.samsung.smarthome.Diagnostics.execute(line 789)
```

Log Reference	LOG-33050

5. Executed Steps at Time of Failure

1	Check server logs for event code 'EVT-1800'.	The performance counter logs a value below the SLA threshold.

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2	Disconnect the hub and wait 5 seconds.	The performance counter logs a value below the SLA threshold.
3	Assert that database record for entity 'ENT-473' has status='PEND...	No crash or ANR is observed.
4	Simulate network drop using the network emulator profile 'Offline...	A confirmation notification appears on the paired device.

6. Root Cause Analysis

Retry logic does not apply exponential back-off; storm of reconnection attempts saturates the gateway.

7. Linked Defect

Defect ID	BUG-0164
Title	Assertion Failure in Sensor Manager: AssertionError: expected '200' but received...
Severity	Trivial
Priority	Critical
Status	FIXED
Root Cause	Retry logic does not apply exponential back-off; storm of reconnection attempts ...
Resolution	Added key rotation broadcast to all replica nodes; integration test verifies pro...
Assigned To	ENG-056

8. Historical Similar Failures

HIST-0071	FAIL-0072	Assertion Failure	5.3.0	2021-02-03	12	Wont Fix
HIST-0354	FAIL-0355	Assertion Failure	6.3.5	2022-05-02	6	Wont Fix
HIST-0522	FAIL-0523	Assertion Failure	4.0.5	2022-01-12	4	Resolved

9. Synthetic Data Notice

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